

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Quiz 1
Duration: 30 minutes

Semester: Fall 2022
Full Marks: 25

CSE 471: System Analysis and Design

Name:	ID:	Section: 03
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1 CO1	Calculate the Break-Even Point and Return of investment using following information.	5															
	<table><tr><td></td><td>Year 0</td><td>Year 1</td><td>Year 2</td><td>Year 3</td></tr><tr><td>Total benefits</td><td>20,000</td><td>35,000</td><td>45,000</td><td>50,000</td></tr><tr><td>Total costs</td><td>75,000</td><td>15,000</td><td>12,000</td><td>20,000</td></tr></table>		Year 0	Year 1	Year 2	Year 3	Total benefits	20,000	35,000	45,000	50,000	Total costs	75,000	15,000	12,000	20,000	
	Year 0	Year 1	Year 2	Year 3													
Total benefits	20,000	35,000	45,000	50,000													
Total costs	75,000	15,000	12,000	20,000													
2 CO1	Suppose a tourism company planning to build a Voice translation app for tourist. Identify 3 functional requirements and 2 non-functional requirements of Voice translation project	5															

3
CO2

Suppose you are the project manager of company X and you are asked to estimate the time required for the project completion. Company X has greater experience working with Java and C#. The following data are provided to you

Measurement Parameter	Low	Average	High
1. Inputs	3	4	6
2. Outputs	4	5	7
3. Queries	3	4	6
4. Files	7	10	15
5. Programming interfaces (EIF)	5	7	10

Table 1: Complexity Index Table

Measurement Parameter	Low	Average	High
1. Inputs	4	2	3
2. Outputs	6	7	10
3. Queries	0	6	6
4. Files	0	16	0
5. Programming interfaces (EIF)	1	0	2

Table 2: System Component Table

Data communications - 3 Heavy use configuration - 3
 Transaction rate - 0 End-user efficiency - 0
 Complex processing - 0 Installation ease - 0
 Multiple sites - 0 Performance - 2
 Distributed functions - 2 Online data entry - 2
 Online update - 2 Reusability - 0
 Operational ease - 0 Extensibility - 0
 (0 = no effect on processing complexity; 3 = great effect on processing complexity)
 Total Processing Complexity (PC):

Language	Approx No. of Lines of code per Function Point
Java	55
C#	50
Python	30

Suggest a suitable Programming Language that would be chosen by you as a manager.

4 CO1	1. Discuss major pros and cons of DevOps (2 of each)	3
5 CO1	Abid runs a supershop with products mainly containing daily needs. Abid has seen a fall in his sales due to the behavioral changes of customer(shifting to online). Abid consults your company to make create an online web and mobile app for his supershop. You are assigned to create a system request containing the project sponsor, business requirement, business value and any special Issue that should be considered. Note that you have already done some market research you are well aware of the figures(Assumptions regarding statistics and Figures are allowed).	6